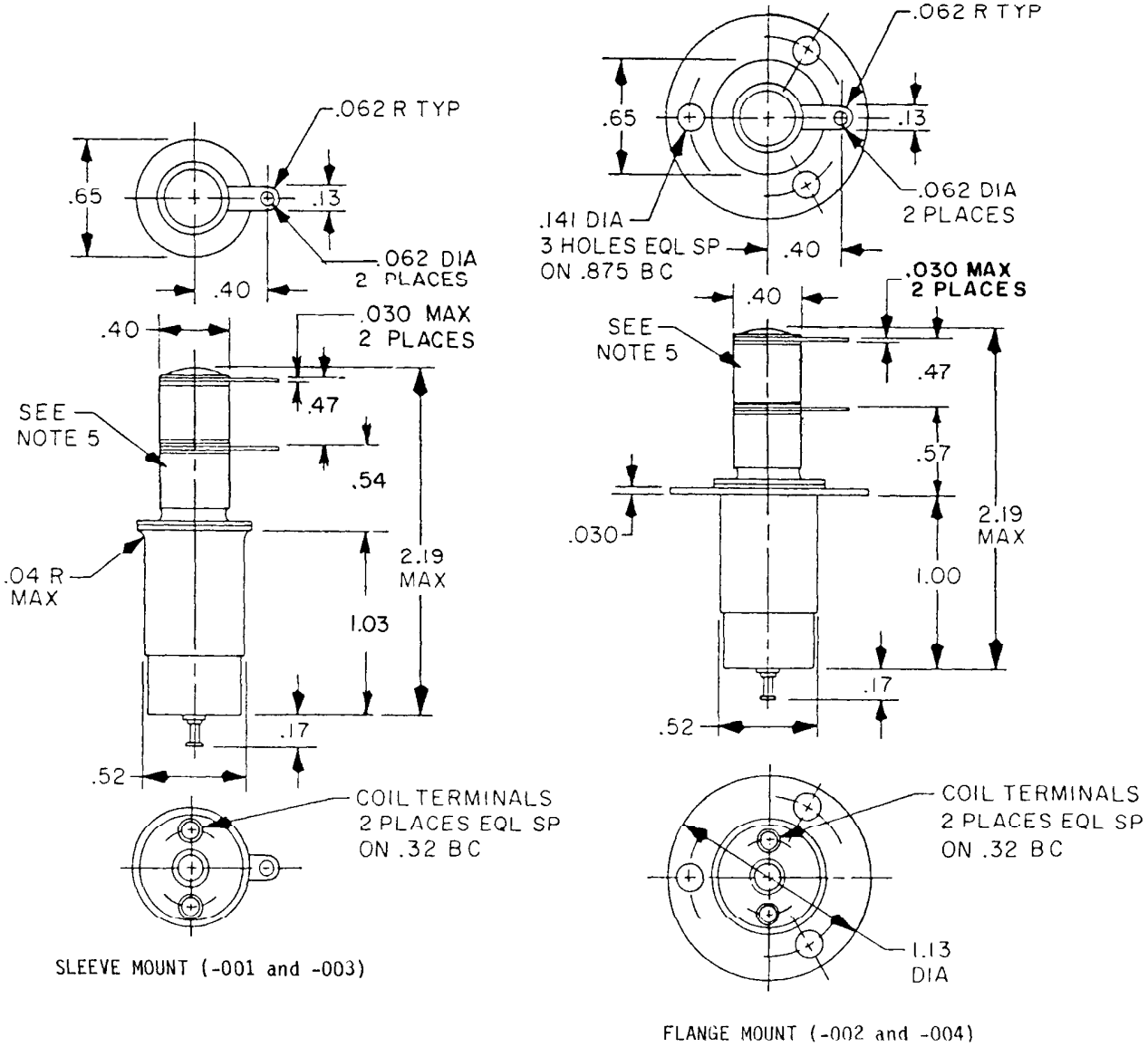
MILITARY SPECIFICATION SHEET

RELAYS, VACUUM, SPST, N/C, 12 AMPERES DC OR 60 Hz RMS, 10 KILOVOLTS DC OR 60 Hz PEAK

This specification sheet is approved for use by the Department of the Air Force, and is available for use within all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-R-83725.



SLEEVE MOUNT (-001 and -003)

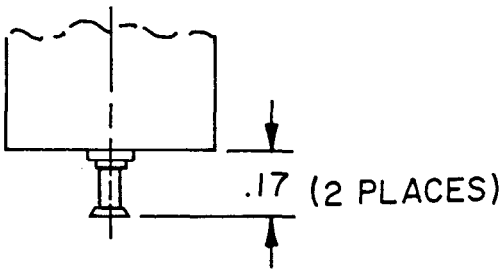
FLANGE MOUNT (-002 and -004)

Ⓐ FIGURE 1. Relay, dimensions and configurations.

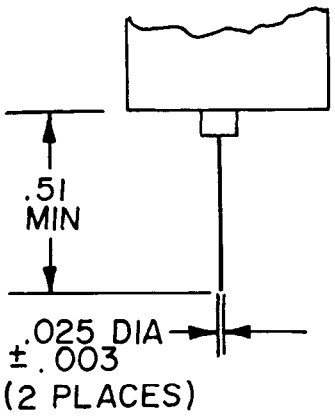
Ⓐ denotes changes.

AMSC N/A
FSC 5945

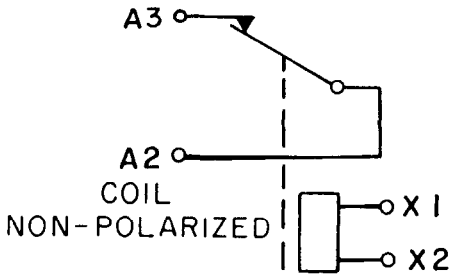
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
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Turret terminal
(see note 4)



Wire lead



CIRCUIT DIAGRAM

Inches	mm	Inches	mm
.015	0.38	0.51	13.0
.025	0.64	0.52	13.2
.03	0.76	0.54	13.7
.04	1.02	0.57	14.5
.062	1.57	0.65	16.5
.13	3.30	0.875	22.2
.141	3.58	1.00	25.4
.17	4.32	1.03	26.2
.32	8.13	1.13	28.7
.40	10.2	2.19	55.5
.47	11.9		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Shape of turret terminals optional, however they shall accommodate two number 22 AWG wires as a minimum.
5. Glazed finish on ceramic insulator is optional.
6. Sleeve mount relays are mounted by .52 inch (13.2 mm) diameter of body of coil housing or by .65 inch (16.5 mm) diameter lip in center of relay.

Ⓐ FIGURE 1. Relay, dimensions and configurations - Continued.

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REQUIREMENTS

CONTACT DATA:

Configuration: SPST, N/C, ground isolated.

Arrangement: 1 form B.

Load ratings:

Resistive: (Carry only).

12 amperes dc or 60 Hz rms.

10 amperes rms, 2.5 MHz.

6 amperes rms, 16 MHz.

Rated operating voltage:

10 kilovolts peak, 60 Hz or dc.

7 kilovolts peak, 2.5 MHz.

6 kilovolts peak, 16 MHz.

Contact resistance:

Rated life:

Before: .020 ohm, maximum.

During: .100 ohm, maximum.

After: .100 ohm, maximum.

Contact bounce: Not applicable.

Capacitance:

1.6 picofarads, maximum across open contacts.

1.6 picofarads, maximum between open contacts and ground.

COIL DATA: (See table I).

Duty rating: Continuous.

Operate time: 15 milliseconds, maximum over temperature range with nominal coil voltage.

Release time: 15 milliseconds, maximum over temperature range from nominal coil voltage.

ELECTRICAL DATA:

Insulation resistance: 1,000 megohms, minimum, except the resistance between coil and housing at high temperature shall be 500 megohms or greater.

Dielectric withstanding voltage:

At atmospheric pressure: 60 Hz.

Between all mated contacts in the open position: 12 kV peak.

Between high voltage terminals and housing: 12 kV peak.

Between coil and housing: 500 V rms, 60 Hz.

Ground isolated: Ground isolated relays have operating elements insulated from the ground plane to provide absolute voltage isolation between the frame and the high voltage contacts.

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ENVIRONMENTAL DATA:

Temperature range: -55°C to +125°C.

Vibration: MIL-STD-202, method 204, test condition C (10 to 55 Hz, .06 inch DA/55 to 2000 Hz, 10 g).

PHYSICAL:

Terminal strength: 5 pounds pull.

Dimensions and configuration: See figure 1.

Ⓐ Terminations: See figure 1.

Weight: 1 ounce, maximum.

LIFE TEST REQUIREMENTS:

Mechanical cycling: 1,000,00 cycles. 2 sample units (cycling rate, 36,000 per hour maximum; contact current shall not exceed 10 milliamperes).

QUALITY ASSURANCE:

Dielectric withstanding voltage:

Tests to be conducted at atmospheric pressure rating only.
Duration of application: 5 to 10 seconds at a 10 percent increase in the dielectric withstanding voltage.

PART NUMBER: See table I.

Ⓐ TABLE I. Dash numbers and characteristics.

Dash number M83725/10-	Coil data (V dc)		At 25°C				Over temperature range			Mounting	Terminal type
	Nominal 1/	Maximum	Coil resistance ohms ±10%	Pickup voltage (V dc)	Hold voltage (V dc)	Dropout voltage (V dc)	Pickup voltage (V dc)	Hold voltage (V dc)	Dropout voltage (V dc)		
-001	26.5	32	290	16	10	1.0	22	14	0.7	Sleeve	Turret
-002	"	"	"	"	"	"	"	"	"	Flange	Turret
-003	"	"	"	"	"	"	"	"	"	Sleeve	Wire
-004	"	"	"	"	"	"	"	"	"	Flange	Wire
-005	12	16	70	8.0	5.0	0.5	11	7.0	0.3	Sleeve	Turret
-006	"	"	"	"	"	"	"	"	"	Flange	Turret
-007	"	"	"	"	"	"	"	"	"	Sleeve	Wire
-008	"	"	"	"	"	"	"	"	"	Flange	Wire

1/ CAUTION: Use of any coil voltage less than the nominal coil voltage will compromise the operation of the relay.

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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